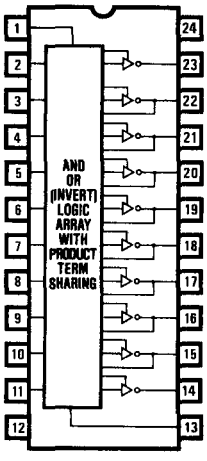
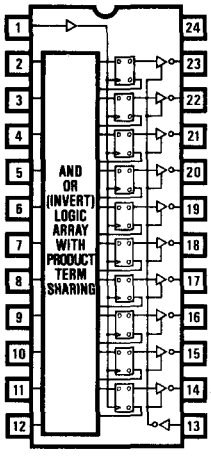
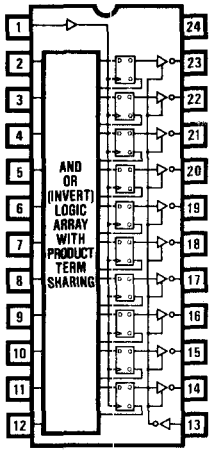
20S10



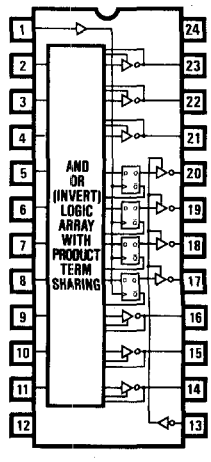
20RS10



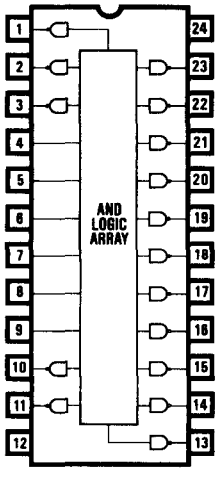
20RS8



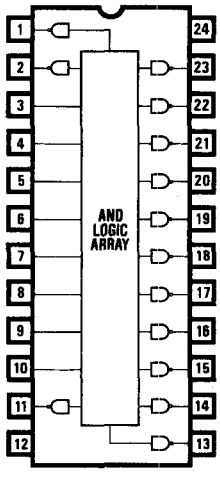
20RS4



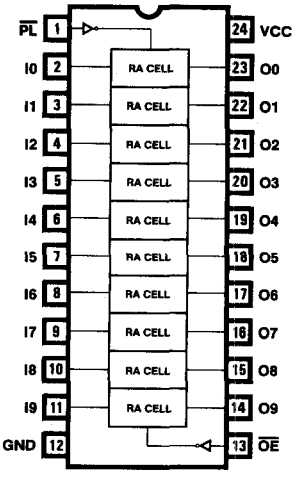
6L16A



8L14A

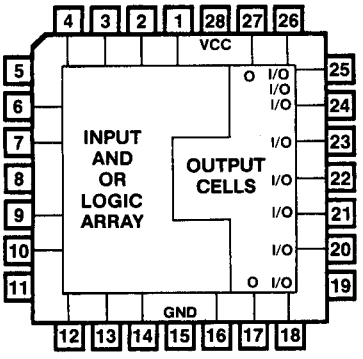


20RA10

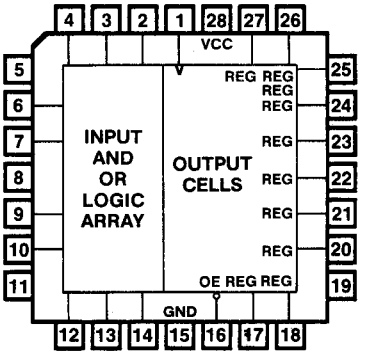


5

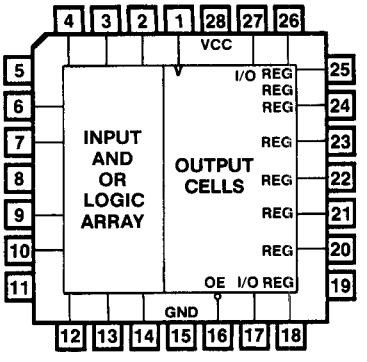
20S10



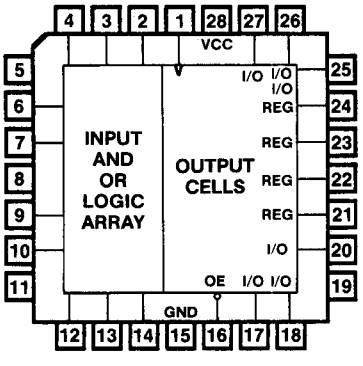
20RS10



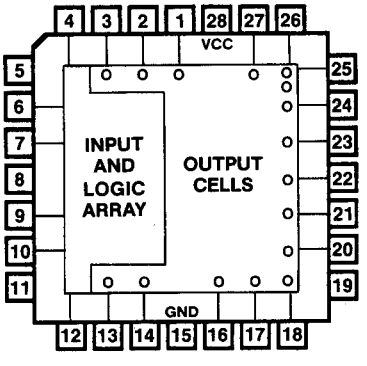
20RS8



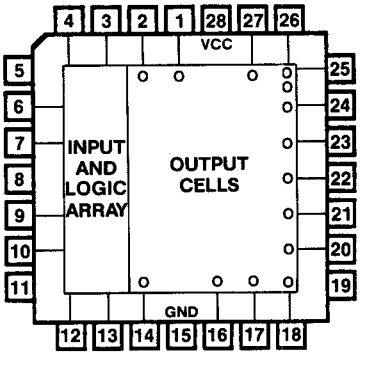
20RS4



6L16A



8L14A



Operating Conditions

SYMBOL	PARAMETER		MILITARY MIN	MILITARY TYP	MILITARY MAX	COMMERCIAL MIN	COMMERCIAL TYP	COMMERCIAL MAX	UNIT
V_{CC}	Supply voltage		4.5	5	5.5	4.75	5	5.25	V
t_w	Width of clock	Low	20	10		15	10		ns
		High	20	10		15	10		
t_{su}	Setup time from input or feedback to clock	20RS10, 20RS8, 20RS4	40	25		35	25		ns
t_h	Hold time		0	−10		0	−10		ns
T_A	Operating free-air temperature		−55			0		75	°C
T_C	Operating case temperature				125				°C

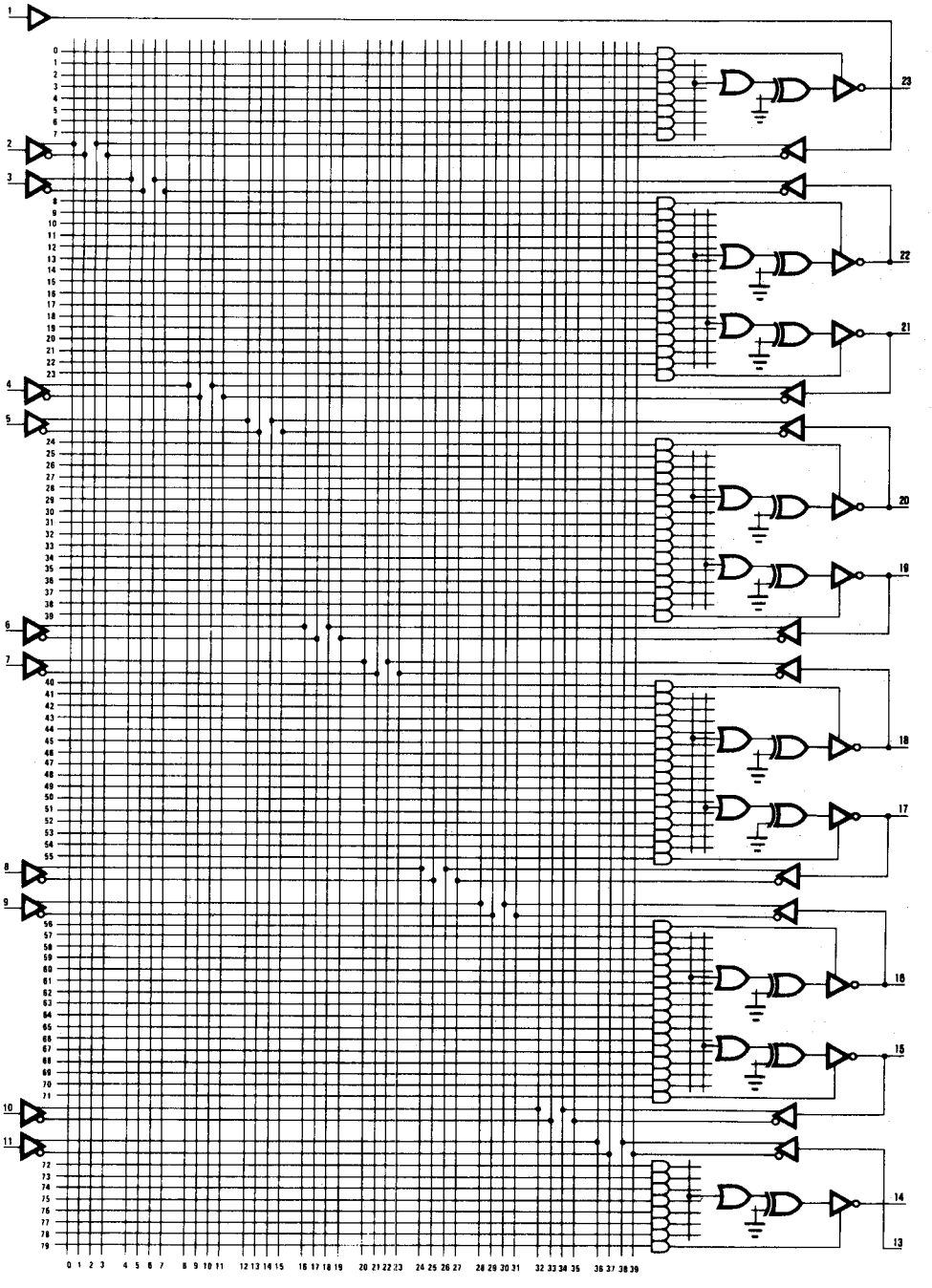
Electrical Characteristics Over Operating Conditions

SYMBOL	PARAMETER	TEST CONDITION			MIN	TYP	MAX	UNIT
V_{IL}*	Low-level input voltage						0.8	V
V_{IH}*	High-level input voltage				2			V
V_{IC}	Input clamp voltage	V_{CC} = MIN	I_I = −18 mA			−0.8	−1.5	V
I_{IL}†	Low-level input current	V_{CC} = MAX	V_I = 0.4 V			−0.02	−0.25	mA
I_{IH}†	High-level input current	V_{CC} = MAX	V_I = 2.4 V				25	μA
I_I	Maximum input current	V_{CC} = MAX	V_I = 5.5 V				1	mA
V_{OL}	Low-level output voltage	V_{CC} = MIN	Mil	I_{OL} = 12 mA		0.3	0.5	V
			Com	I_{OL} = 24 mA				
V_{OH}	High-level output voltage	V_{CC} = MIN	Mil	I_{OH} = −2 mA	2.4	2.8		V
			Com	I_{OH} = −3.2 mA				
I_{OZL}†	Off-state output current	V_{CC} = MAX		V_O = 0.4 V			−100	μA
I_{OZH}†				V_O = 2.4 mA			100	
I_{OS}**	Output short-circuit current	V_{CC} = 5 V		V_O = 0 V	−30	−70	−130	mA
I_{CC}	Supply current	V_{CC} = MAX				175	240	mA

Switching Characteristics Over Operating Conditions

SYMBOL	PARAMETER		TEST CONDITIONS	MILITARY MIN	MILITARY TYP	MILITARY MAX	COMMERCIAL MIN	COMMERCIAL TYP	COMMERCIAL MAX	UNIT
t_{PD}	Input or feedback to output 20S10, 20RS8, 20RS4	Polarity fuse intact	Commercial R_1 = 200 Ω R_2 = 390 KΩ		25	40		25	35	ns
		Polarity fuse blown			30	45		30	40	
t_{CLK}	Clock to output or feedback				12	20		12	17	ns
t_{PZX}	Pin 13 to output enable except 20S10				10	25		10	20	ns
t_{PXZ}	Pin 13 to output disable except 20S10				11	25		11	20	ns
t_{PZX}	Input to output enable	20S10, 20RS8, 20RS4			25	35		25	35	ns
t_{PXZ}	Input to output disable	20S10, 20RS8 20RP4	Military R_1 = 390 Ω R_2 = 750 Ω		13	30		13	25	ns
f_{MAX}	Maximum frequency	20RS10, 20RS8, 20RS4		18	28		20	28		MHz

20S10


1
2
3
4
5
6
7
8
9
10
11
23
22
21
20
19
18
17
16
15
14
13
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39